
COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

LAB REPORT

Lab Title:

Development of UET Relative GPA Management System Using C++

1. OBJECTIVE

The objective of this lab is to design and implement a console-based **Relative GPA Management System** using the C++ programming language. The project enables the management of student academic records by providing facilities to add student information, enter examination marks, calculate relative GPA, generate student rankings, display class statistics, and save or load student data using file handling. The project also strengthens understanding of object-oriented programming concepts, arrays, structures, functions, and file management.

2. THEORY

A Student Management System is an important application used by educational institutions to organize and maintain student academic records efficiently. In this project, a Relative GPA system has been developed following the grading policy where student grades are assigned relative to the overall class performance rather than fixed percentage boundaries.

The program is written in **C++**, an object-oriented programming language that supports modular programming through classes, structures, functions, and file handling. The system stores student information such as roll number, name, department, batch, and examination marks.

A **Student** class is used to store and manage student records, while a **Marks** structure stores quiz, assignment, midterm, and final examination marks. Arrays are used to maintain multiple student records. Functions are created for each system operation to improve modularity and readability.

Relative GPA is calculated by first determining the class average. Students are then assigned grades according to their performance relative to the average. The program also calculates class statistics including highest marks, lowest marks, and average marks.

To ensure permanent storage of records, **file handling** is implemented using input and output file streams. Student records can be saved into a text file and loaded again whenever the program is executed.

3. SOFTWARE USED

- DEV-C++ IDE
 - C++11 Compiler
 - Windows Operating System
 - Text File (students.txt) for data storage
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4. PROCEDURE

Step 1 – Create a New Project

- Open DEV-C++.
- Create a new C++ source file.
- Save the project as **UET Relative GPA System.cpp**.

Step 2 – Include Required Libraries

Include the following header files:

- `iostream`
- `fstream`
- `string`

These libraries provide input/output operations, file handling, and string manipulation.

Step 3 – Create Student Data Structure

Create a **Marks** structure to store:

- Quiz Marks
- Assignment Marks

- Mid Marks
- Final Exam Marks

Create a **Student** class containing:

- Roll Number
 - Student Name
 - Department
 - Batch
 - Marks
 - Total Marks
 - GPA
 - Grade
-

Step 4 – Implement Student Functions

Develop functions to perform the following operations:

- Add Student
 - Display Student Records
 - Search Student
 - Enter Student Marks
 - Calculate Relative GPA
 - Display Statistics
 - Rank Students
 - Save Student Data
 - Load Student Data
-

Step 5 – Calculate Relative GPA

The program first calculates the class average marks.

Grades are assigned according to the following conditions:

Condition	Grade	GPA
Average + 15 or above	A	4.00
Average + 10	B	3.50
Average to Average +10	C	3.00
Average -10 to Average	D	2.00
Below Average -10	F	0.00

Step 6 – Display Statistics

The system calculates:

- Highest Marks
- Lowest Marks
- Average Marks

These statistics help analyze the overall class performance.

Step 7 – Rank Students

Students are sorted in descending order of total marks using the Bubble Sort algorithm.

The ranking displays:

- Rank
 - Student Name
 - Roll Number
 - Total Marks
 - GPA
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Step 8 – Save and Load Data

Student information is stored in a file named:

students.txt

Whenever required, the program can reload all saved records from the file.

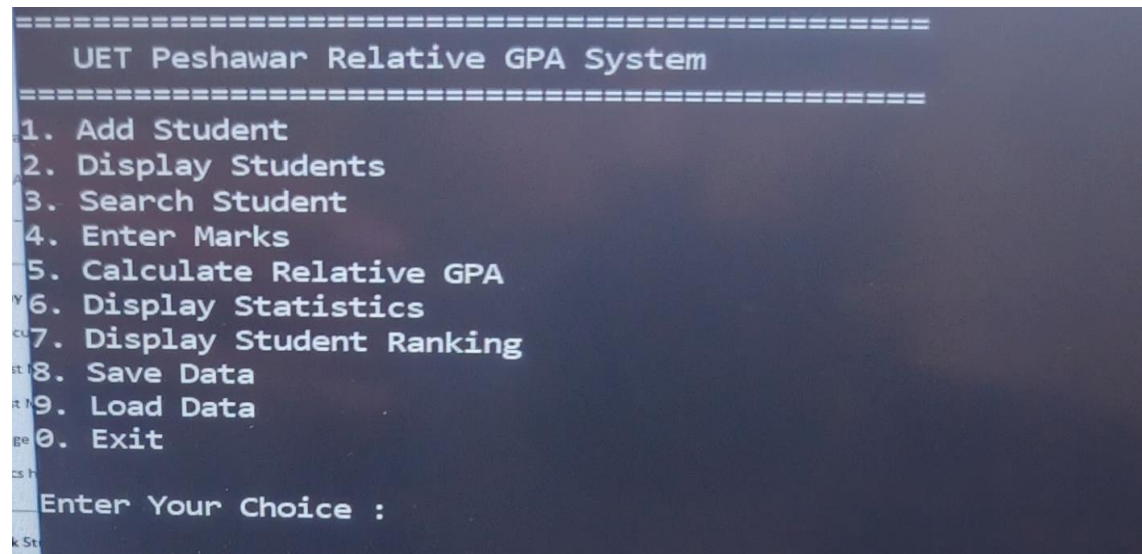
Step 9 – Compile and Run

Compile the program.

Execute the program.

The following menu appears:

```
=====
                        UET Peshawar Relative GPA System
=====
```

A screenshot of a terminal window displaying the main menu of the 'UET Peshawar Relative GPA System'. The menu is enclosed in a dashed border and lists ten options: 1. Add Student, 2. Display Students, 3. Search Student, 4. Enter Marks, 5. Calculate Relative GPA, 6. Display Statistics, 7. Display Student Ranking, 8. Save Data, 9. Load Data, and 0. Exit. Below the list, it prompts the user to 'Enter Your Choice :'.

```
=====
                        UET Peshawar Relative GPA System
=====
1. Add Student
2. Display Students
3. Search Student
4. Enter Marks
5. Calculate Relative GPA
6. Display Statistics
7. Display Student Ranking
8. Save Data
9. Load Data
0. Exit
Enter Your Choice :
```

5. PROGRAM FEATURES

The developed system provides the following features:

- Add new student records.

- Store complete student information.
- Enter quiz, assignment, midterm, and final marks.
- Calculate total marks automatically.
- Calculate Relative GPA.
- Assign grades according to class average.
- Display complete student details.
- Search students using roll number.
- Display class statistics.
- Rank students according to performance.
- Save records permanently using file handling.
- Load previously saved student records.
- User-friendly menu-driven interface.

6. SAMPLE OUTPUT

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UET Peshawar Relative GPA System

=====

1. Add Student
2. Display Students
3. Search Student
4. Enter Marks
5. Calculate Relative GPA
6. Display Statistics
7. Display Student Ranking

8. Save Data

9. Load Data

0. Exit

Enter Your Choice: 1

Enter Roll Number : 23PWEE001

Enter Student Name : Ali Khan

Enter Department : Electrical Engineering

Enter Batch : 2

Student Added Successfully.

After GPA calculation:

Class Average Marks : 73.4

Relative GPA Calculated Successfully.

Statistics:

Highest Marks : 94

Lowest Marks : 52

Average Marks : 73.4

Ranking:

Rank 1

Name : Ali Khan

Roll : 23PWEE001

Marks : 94

GPA : 4.00

7. CONCLUSION

The Relative GPA Management System was successfully designed and implemented using C++. The project demonstrated the practical use of object-oriented programming concepts such as classes, structures, arrays, functions, and file handling. The system efficiently manages student records, calculates relative GPA based on class average, ranks students according to performance, and stores records permanently using text files. This project enhanced understanding of modular programming, data management, sorting algorithms, and file operations, providing valuable experience in developing real-world academic management applications.